

KRISH BANSAL

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EDUCATION

Virginia Tech | Blacksburg, VA

Expected May 2027

Bachelor of Science in Computer Science

Cumulative GPA: 3.62 / 4.0 | **In Major GPA:** 3.68 / 4.0

Dean's List with Distinction: Spring 2025, Fall 2025 | **Dean's List:** Fall 2023, Fall 2024, Spring 2026

PROJECTS EXPERIENCE

- Change Assurance Runtime for AI Coding Agents** | *VT Hacks 14* | *Python, FastAPI, Textual, Electron/React, SQLite* Sep 2026
- Built a local first Windows runtime that puts AI coding agents under independent supervision, using kill on close Job Objects and restricted token process launches to attribute every descendant process by PID, image path, and command line
 - Designed a tamper evident, hash chained event journal with end to end replay verification and a signed Ed25519 Change Passport capturing intent, authority, evidence, and outcomes for every agent action
 - Shipped three clients (Textual TUI, CLI, Electron/React desktop) against one frozen OpenAPI contract of 66 operations, 61 routes, and 110 typed schemas, validated by a threat model review (15 of 16 findings fixed) and 1,022 passing tests
- Keshav Enterprises | Intern | India** | *Python, PostgreSQL, FastAPI, Docker, MLflow* May 2026- Aug 2026
- Architected end-to-end ELT pipeline in Python and PostgreSQL, normalizing sales inquiry and order data across 4 relational entities to replace manual spreadsheet tracking for an SME manufacturer
 - Built a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB vector search and the Claude API to auto-draft grounded customer replies, constraining outputs to retrieved catalog data to eliminate hallucinated pricing
 - Designed and ran an ML evaluation harness benchmarking Prophet time-series forecasts against a naive seasonal baseline across 8 product lines, using MLflow to gate model deployment – correctly rejected the more complex model in 7/8 cases due to limited (24-month) historical data, preventing an unjustified model from shipping
 - Achieved a 49% relative MAPE reduction (52.0% → 26.6%) with Prophet on the one product line where sufficient trend signal existed, validating the evaluation harness caught a genuine case for deployment
- Regime Detection & Adaptive Trading Strategies** | *Python, pandas, NumPy, hmmlearn, scikit-learn* March 2026 – May 2026
- Built a Gaussian Hidden Markov Model (HMM) in Python to detect market regimes from SPY/VIX data using volatility, momentum, liquidity, and volume features
 - Developed a transaction-cost-aware backtesting framework with filtered probability inference and benchmarked against K-Means and momentum strategies, achieving a **57.97%** return with **~34%** lower volatility than Buy-and-Hold
- Fork-Join Thread Pool** | *CS 3214* | *C, pthreads, Helgrind, Git* Spring 2026
- Built a work-stealing fork/join thread pool in C with pthreads, evolving a single-lock work-sharing queue into per-worker dequeues with finer-grained locking so exactly one thread runs each task
 - Prevented deadlock by having workers run not-yet-started joined tasks inline, kept idle workers blocked on condition variables, checked for data races with Helgrind, and benchmarked parallel speedup on 32-core Xeon machines
- Undergraduate Research Assistant | Dataism Laboratory for Quantitative Finance** Oct 2025 – Aug 2026
- Built a CNN-Transformer **hybrid model** for high-frequency market data, using convolutional filters to extract short-horizon trading signals and transformer attention to model long-range temporal dependencies for price direction forecasting using **Python, R**
- Real Estate Analyzer** | *VT Hacks* | *ReactJS, NodeJS, PostgreSQL, Azure ML Studio* Oct 2024 – Nov 2024
- Built a Boosted Decision Tree model predicting real estate prices with **97.5% accuracy ($R^2 = 0.975$)** on **428K+ listings across 45+ U.S. states**, reducing **MAE to \$87.9K** and **RMSE to \$181.8K** via hyperparameter tuning and feature engineering in **Azure ML Studio**
 - Delivered a full-stack web prototype at VTHacks 12, integrating the model with **React.js, Node.js, and PostgreSQL** to enable scalable real estate price analysis
- Playlist Recommendation** | *Java, SQL* Feb 2024 – May 2024
- Engineered a personalized playlist recommendation system in **Java** using hybrid filtering, processing **10K+ songs** and **1K+ user preference profiles** to generate ranked recommendations
 - Improved recommendation relevance by **35%** through dynamic weighting and similarity scoring, reducing average selection time and increasing user engagement
- Inventory Management System** | *Java, SQL* Aug 2023 – Oct 2023
- Architected and implemented a scalable inventory management system in **Java** with a **relational SQL backend**, supporting real-time stock tracking, automated restocking logic, and invoice generation, improving operational efficiency by **30%**
 - Designed normalized database schemas and modular object-oriented architecture (OOP), optimizing query performance and ensuring data integrity while documenting system design and scalability in a detailed **Technical Design Document (TDD)**
- Uber | Intern** | *Remote* Oct 2022 – Dec 2022
- Performed data-driven market analysis and competitive benchmarking to identify user segments and feature opportunities for the **"Uber-Teens"** prototype, and designed high-fidelity **UI/UX mockups**, optimizing user flows and interaction design while presenting product strategy insights to Uber stakeholders and industry mentors

TECHNICAL SKILLS

Languages: Java, Python, Swift, C, C++, JavaScript, TypeScript, SQL, R, Ruby

Frameworks: FastAPI, React.js, Next.js, Node.js, Ruby on Rails, Electron, Scikit-learn, spaCy, Bootstrap

Libraries: Pydantic, pandas, NumPy, Matplotlib, TensorFlow, PyTorch, Prophet, hmmlearn

Databases: PostgreSQL, SQLite, ChromaDB, SQL

Developer Tools: Git, Docker, MLflow, Azure ML Studio, Google Cloud Platform, Figma